Investigating the Use of Edge Computing for Improving IoT Applications

Edge computing is a technology that allows data processing to occur closer to the source of the data, rather than in a centralized location such as a data centre. This can be particularly useful for Internet of Things (IoT) applications, as it allows for faster response times and reduced latency.

When investigating the use of edge computing for improving IoT applications, there are several key areas to consider:

1. **Data processing and storage:**
   Edge computing can enable real-time processing of data from IoT devices, allowing for faster decision making and response times. It can also reduce the amount of data that needs to be sent to a centralized location for processing, reducing the load on networks and improving scalability.
2. **Security:**
   Edge computing can provide an additional layer of security for IoT applications by allowing for the processing and storage of sensitive data closer to the source. This can help to prevent data breaches and protect against cyber threats.
3. **Network connectivity:**
   Edge computing can improve the reliability and availability of IoT applications by reducing the dependence on centralized infrastructure. This can be particularly beneficial in areas with limited or unreliable network connectivity.
4. **Cost:**
   Edge computing can help to reduce costs associated with IoT applications by reducing the amount of data that needs to be sent to a centralized location for processing. It can also help to reduce the need for expensive specialized hardware and software.
5. **Flexibility:**
   Edge computing can be easily adapted to different IoT use cases and can be implemented at the edge of the network, close to the devices and sensors. This allows for a more flexible and responsive IoT ecosystem.
6. **Use cases:**
   Edge computing is beneficial for IoT use cases that require low latency and high throughput, such as autonomous vehicles, industrial control systems, and smart buildings.

To investigate the use of edge computing for improving IoT applications, it is important to conduct research on the specific requirements of the application and the technology options available. This can include conducting feasibility studies, evaluating existing solutions, and conducting proof-of-concept testing. Additionally, it is important to consider the scalability, security and cost-effectiveness of the solution.

Overall, Edge computing can help to improve the performance and reliability of IoT applications, while reducing costs and improving security. It is a technology that is well worth investigating for organizations looking to take advantage of the opportunities provided by IoT.